BASE CHANGE ALONG A SURJECTIVE FLAT MORPHISM IS FAITHFUL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays a commutative diagram relating the two morphisms and their base changes; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 3 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- A surjective flat morphism of schemes is an epimorphism. Same chapter, Lemma lemma-surjective-flat-epi, invoked at proof lines 8753-8754. Provides the cancellation property that gives faithfulness.
- Surjectivity and flatness are preserved by base change. Morphisms, Lemma morphisms-lemma-base-change-surjective at proof lines 8751-8753 and Morphisms, Lemma morphisms-lemma-base-change-flat at proof lines 8751-8753. Shows that the projection whose being an epimorphism is needed satisfies the hypotheses of the epimorphism lemma.

Source reference. Original lemma 02VW: sources/stacks-descent/pinned-original-source.tex, source lines 8700–8700. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01U9: sources/stacks-chow/tags.txt, source lines 2402–2402. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01S1: sources/stacks-chow/tags.txt, source lines 2322–2322. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $h : S' \to S$ be a surjective, flat morphism of schemes. The base change functor*

$$Sch/S \longrightarrow Sch/S', \quad X \longmapsto S' \times_S X$$

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is faithful.

Proof. Let X_1, X_2 be schemes over S. Let $\alpha, \beta : X_2 \to X_1$ be morphisms over S. If α, β base change to the same morphism then we get a commutative diagram as follows

$$\begin{array}{ccccc} X_2 & \longleftarrow & S' \times_S X_2 & \longrightarrow & X_2 \\ \downarrow{\scriptstyle \alpha} & & \downarrow & & \downarrow{\scriptstyle \beta} \\ X_1 & \longleftarrow & S' \times_S X_1 & \longrightarrow & X_1 \end{array}$$

Hence it suffices to show that $S' \times_S X_2 \to X_2$ is an epimorphism. As the base change of a surjective and flat morphism it is surjective and flat (see Morphisms, Lemmas 01S1 and 01U9). Hence the lemma follows from Lemma 02VW. $\square$